

SAIC-PV's Smart Ningde Factory

**1 Car /min
4 Workshops
5 Car Platforms
10 Models Production lines**



**Digital Transformation Based on Huawei
Intelligent IP Networks**

A Digital Upgrade For Chinese Manufacturing

A steady push for the long

As China's digital transformation gains huge momentum, "Four New Great Inventions": High-speed rail, E-commerce, Mobile payment and Bike sharing have shaped all aspects of people's life. However, how to fully leverage the changes and opportunities of a mix of digital technologies and their accelerating impact across all sectors in a strategic and prioritized way features prominently in China's digital transformation efforts.

**Building a
Intelligent
world**



Disruptive Trends in the Auto Industry

SAIC's Ningde Factory sets a good example

The disruptive impacts of digital revolution is greatly felt in the auto industry, which has a long industrial chain.

The car production involves many processes: Stamping – Welding – Painting – Assembling – Inspection and so on. It also calls for close collaboration among more

than 100 sectors, including the steel industry, the oil industry, the chemical industry and the electronics industry. As such, the digital upgrade of the automotive industry will have a ripple effect on every link of the value chain, including manufacturing, warehousing, and distribution.

The most pressing challenge for auto companies is how fast and far they can go on their path to digital transformation.

In this regard, Ningde factory gets the ahead of game.



SAIC's Smart Ningde Factory

“1+4” SAIC: Digital Transformation Strategy

1

Digital Products

-Intelligent Connected Cars

4

Digital Marketing

-Precise and insightful customer portraits
-The next level user experience

Intelligent Manufacturing

-Industrial Interconnection
-Big data Intergration

Digital R&D

-Software-defined hardware

Smart Industrial Park

-Flexibility & Collaboration

Flexibility Digitalization Automation Data-driven

Supported by stable, flexible and secure ICT infrastructure products, SAIC's Ningde base works to bring digital to all factors and the whole system for a fully connected, intelligent factory.

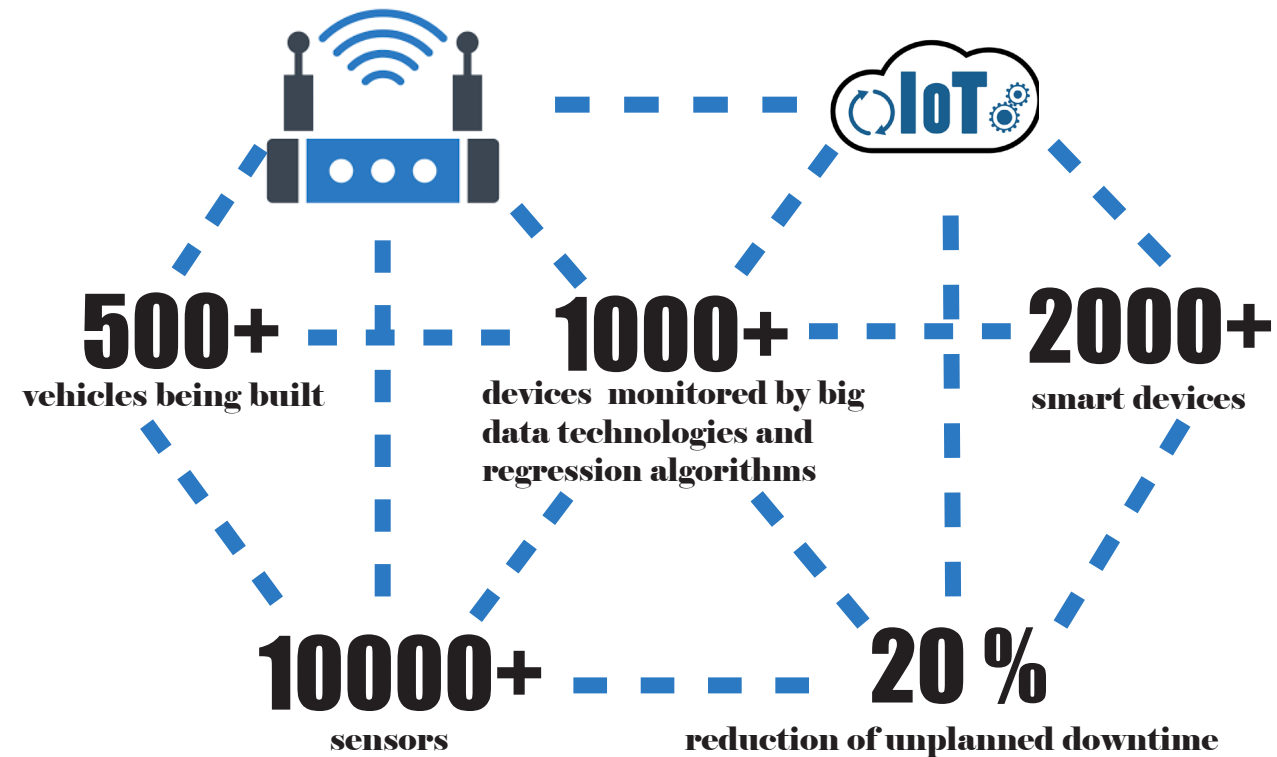
It has accelerated the transformation towards intelligent connectivity in manufacturing processes, application scenarios, a large amount of data, working areas and human resources.

Thanks to intelligent IP solutions Huawei has to offer, SAIC's Ningde factory has led the automotive industry in building more resilient ICT infrastructure, pushing forward with intelligent manufacturing and making innovations through digital collaboration.

Intelligent WORKSHOP



Huawei Agile Campus Network Solution



Working with an IoT expansion module, Huawei's leading AP effectively connects the Ningde factory where has 511 robots. Multi-network integration and intelligent interference avoidance technologies can effectively improve network application efficiency.



FACTS ABOUT THE SMART INDUSTRIAL PARK

BASED ON HUAWEI INTELLIGENT IP NETWORKS

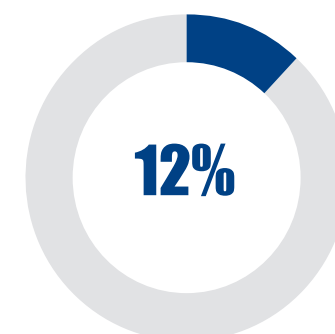
Huawei's next-generation AP

Ultra-low Latency
High Throughput
Self-Organizing Networking

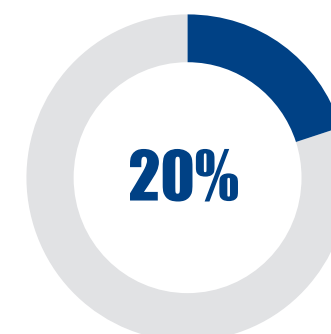
High quality video conferencing
Rapid cross-base data transfer

Careful parts inventory management
Efficient manufacturing supply

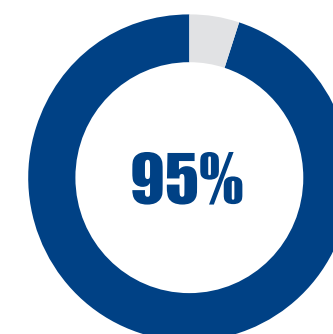
Refined warehouse management
Economical distribution strategy



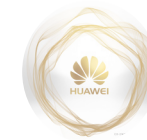
Reduction in the area of on-site logistics



Increase in coverage: Huawei's cutting edge wireless AP that uses smart antenna technology



On-time Delivery





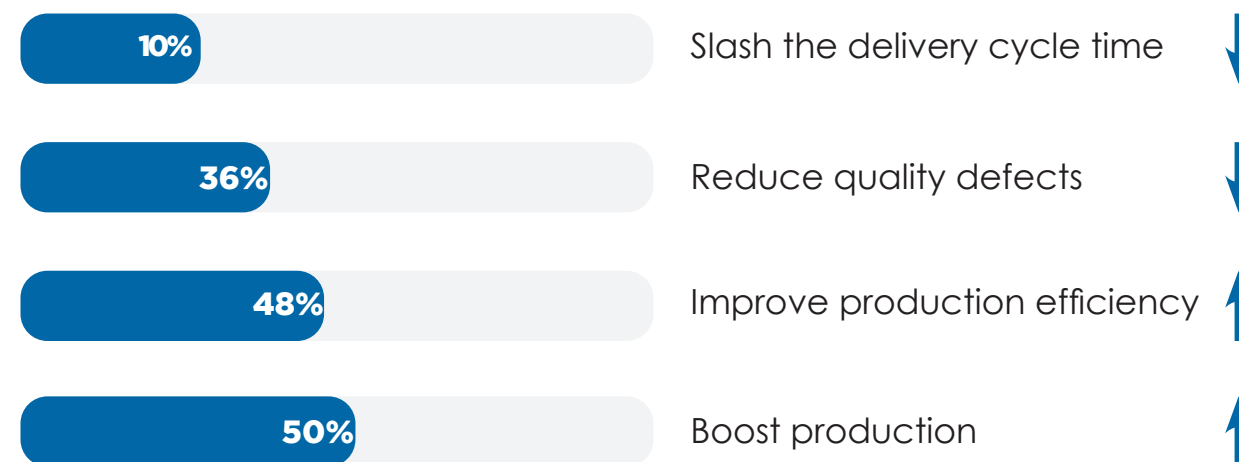
2020 Model Smart Factory Award

AI-enabled Intelligent Network Operations & Maintenance

ultra-high bandwidth ultra-low latency massive connections

Huawei 5G networks help SAIC's Ningde base:

- Deal with its complex business portfolios;
- Handle the surge in internet traffic;
- Navigate massive amounts of production data;
- Promote the application of Automated Guided Vehicles (AGV);
- Facilitate data transfer.



ONE BASIC

Interconnection

TWO PLATFORMS

The industrial big data platform

The industrial IoT platform

TWO TECHNOLOGY

SOLUTIONS

Big data&AI

SAIC's Ningde Factory has achieved

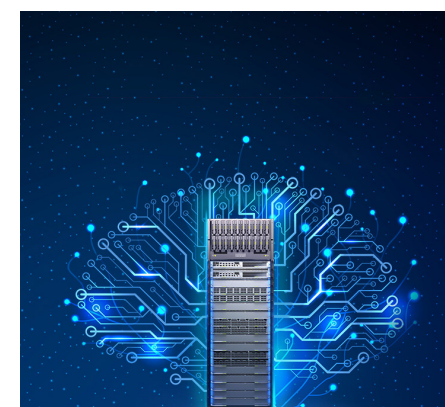
An overall display of production and operation indicators and risks;

Dynamic perception;

Supported decision-making;

A sound intelligent early warning system.

How Huawei Solutions Work



CloudEngine Data Center Switches

High-performance core switches are designed for data center networks providing stable, reliable, secure, and high-performance switching services.



HiSecEngine USG6500E Series

The HiSecEngine USG6500E series can interwork with CloudEngine and other security devices to proactively defend against network threats, enhance border detection capabilities, effectively defend against advanced threats, and resolve performance deterioration problems.



iMaster NCE Autonomous Network Management and Control System

iMaster NCE is a network automation and intelligence platform that integrates management, analysis, and AI functions. It helps Ningde base upgrade operation and maintenance, conduct real-time monitoring, full-life cycle automation, and intelligent closed-loop management.



Looking forward

HUAWEI VISION

Intelligent IP Networks

Super Capacity, Intelligent Experience,
Autonomous Driving



The success story of SAIC's Ningde factory outlines a blueprint for the sustainable development of the automotive industry, while also makes China's automobile industry more competitive internationally. At Huawei, we aim to move our intelligent IP network to the next level as a way of pushing the digital transformation of manufacturing enterprises. Looking forward, Huawei will work with more market players in tailored and innovative design, and renew our commitments to leading the intelligent IP network development for a fully connected, intelligent world.

